Document Identifier: **PMC8844799**

# SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

**Table 1: Rigor Adherence Table**

| |
|---|
| Ethics |
| not required. |
| Inclusion and Exclusion Criteria |
| not required. |
| Attrition |
| not required. |
| Sex as a biological variable |
| not required. |
| Subject Demographics |
| Age: not required. |
| Weight: not required. |
| Randomization |
| not detected. |
| Blinding |
| not detected. |
| Power Analysis |
| not detected. |
| Replication |
| All the samples were assayed in triplicate. |
| Number: All the samples were assayed in triplicate. |

**Table 2: Key Resources Table**

| Your Sentences | REAGENT or RESOURCE | SOURCE | IDENTIFIER |
|---|---|---|---|
| Antibodies | | | |
| They were then treated with PE-labeled CD133 antibody (Miltenyi Biotec, Bergisch Gladbach, Germany) at 4°C for 30min. | CD133 | | |
| The samples incubated with the corresponding control IgG antibody were applied to set the gating level. | control IgG | | |
| Each sample was treated with IgG antibody ( 1:300 , Abcam , Cambridge , MA , USA ) or E2F1 antibody ( 1:200 , Abcam ) for 1h . | IgG | | |
| | E2F1 | | |
| The primary antibodies included: anti-CD133 ( 130–092–395 , 1:100 , Miltenyi Biotec) , anti-CD151 ( SAB1402716 , 1:50 , Sigma) , anti-E2F1 ( ab4070 , 1:500; Abcam) , anti-MDR1 ( ab170904 , 1:100 , Abcam) , anti-BCRP1 ( ab207732 , 1:100 , Abcam) , anti-γ-H2AX ( ab81299 , 1:400; Abcam) , anti-H2AX ( ab11175 , 1:200 , Abcam) , anti-AKT ( #9272 , 1:1000 | anti-CD151 | | |
| | SAB1402716 | | |
| | anti-γ-H2AX | | |
| | anti-H2AX | | |
| | anti-AKT | | |
| | #9272 | | |
| Recombinant DNA | | | |
| For the knockdown of DLGAP1-AS2, specific short hairpin RNA (shRNA) targeting DLGAP1-AS2 was constructed using the pGPU6-GFP-Neo gene vector. | pGPU6-GFP-Neo | | |
| Wild-type ( WT ) and mutant ( MUT ) CD151 sequences containing the predicted E2F1 binding sites were cloned into psiCHECK-2 luciferase reporter plasmid ( Promega , Madison , WI , USA ) to produce luciferase reporter systems , including CD151-WT , CD151-S1 ( MUT2&3) , CD151-S2 ( MUT1&3 ) or CD151-S3 ( MUT1&2) . | psiCHECK-2 | | |
| Software and Algorithms | | | |
| The cells were cultured in RPMI1640 medium (Thermo Fisher, Waltham, MA, USA) with 10% FBS, penicillin (100U/mL, Sigma Aldrich, St. Louis, MO, USA) and streptomycin (100µg/mL, Sigma). | Sigma Aldrich | | Suggestion: Sigma-Aldrich (RRID:SCR_008988)( link) |
| The wound healing area was analyzed using ImageJ software ( National | ImageJ | | Suggestion: ImageJ, RRID:SCR_003070( link) |

| | | | |
|---|---|---|---|
| Institutes of Health , Bethesda , MA , USA) . | | | |
| Dual luciferase reporter gene assay The JASPAR was used for predicting the putative E2F1 binding sites in the CD151 promoter . | JASPAR | | Suggestion: JASPAR (RRID:SCR_003030)( link) |
| The primary antibodies included: anti-CD133 ( 130–092–395 , 1:100 , Miltenyi Biotec) , anti-CD151 ( SAB1402716 , 1:50 , Sigma) , anti-E2F1 ( ab4070 , 1:500; Abcam) , anti-MDR1 ( ab170904 , 1:100 , Abcam) , anti-BCRP1 ( ab207732 , 1:100 , Abcam) , anti-γ-H2AX ( ab81299 , 1:400; Abcam) , anti-H2AX ( ab11175 , 1:200 , Abcam) , anti-AKT ( #9272 , 1:1000 | Sigma | | |
| Statistical analyses were carried out by SPSS 20.0. | SPSS | | Suggestion: SPSS, RRID:SCR_002865( link) |

**Other Entities Detected**

| Your Sentences | Recognized Entity |
|---|---|
| Statistical Tests | |
| The data were analyzed by unpaired two-tailed Student's t-test (two groups) and one-way analysis of variance (multiple groups), followed by Tukey's post-test. | t-test |
| | analysis of variance |
| | Tukey's post-test |
| Oligonucleotides | |
| The sequences for constructing the plasmids of shRNA were showed following: sh-NC: 5#-GTTCTCCGAACGTGTCACGT-3#, sh-DLGAP1-AS2: 5#-GCAAAATTAAAGATTTAGAAG -3#. | 5#-GTTCTCCGAACGTGTCACGT- 3# |
| | 5#-GCAAAATTAAAGATTTAGAAG -3# |
| The primer sequences were presented: DLGAP1-AS2: F 5#-ACATCGTGGCTGAATGAACA- 3# , R 5#-ATCAGTGGGGAGGAAGGAGT-3#; DLGAP1: F 5#-GCTAAATGCTCTCAGCTGCAC –3# , R 5#- GCGCAGCAACTTTTGGAGAA –3#; E2F1: F 5#-ATGTTTTCCTGTGCCCTGAG- 3# , R 5#-AGATGATGGTGGTGGTGAC-3 #; CD151: F 5#-ACAACTGCTGCTTCTGGCTG- 3# , R 5#-AGCAGGATGAAGTACAAGCG-3#; GAPDH: F 5#-TAGATGACACCCGTCCCTGA- 3# , R 5#-ACCTCCACCTGTCCTTAGTG –3# . | 5#-ACATCGTGGCTGAATGAACA- 3# |
| | 5#-ATCAGTGGGGAGGAAGGAGT- 3# |
| | 5#- GCTAAATGCTCTCAGCTGCAC –3# |
| | 5#- GCGCAGCAACTTTTGGAGAA –3# |
| | 5#-ATGTTTTCCTGTGCCCTGAG- 3# |
| | 5#-AGATGATGGTGGTGGTGAC-3 # |
| | 5#-ACAACTGCTGCTTCTGGCTG- 3# |
| | 5#-AGCAGGATGAAGTACAAGCG- 3# |
| | 5#-TAGATGACACCCGTCCCTGA- 3# |
| | 5#-ACCTCCACCTGTCCTTAGTG –3# |

SciScore is an automated tool that is designed to assist expert reviewers by finding and presenting formulaic information scattered throughout a paper in a standard, easy to digest format. ***SciScore is not a substitute for expert review.*** SciScore also checks for the presence and correctness of several unique identifiers, including RRIDs (research resource identifiers) in the manuscript, detects sentences that appear to be missing RRIDs, and can even suggest RRIDs under certain circumstances. **All RRID suggestions should be verified;** only the author can know whether the suggestions are correct.

For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq